

# Asymptotic Normality of the PPCA Loading Matrix in the Quotient Space under High-Dimensional Asymptotics: a Synthesis and Partial Results

Working notes

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## Abstract

We combine three lines of work: (i) the high-dimensional random-matrix analysis of probabilistic principal component analysis (PPCA) of Passemier, Li and Yao (2017), who identified and corrected the bias of the maximum-likelihood noise-variance estimator when  $p/n \rightarrow c > 0$ ; (ii) the quotient-space resolution of the rotational non-identifiability of the PPCA loading matrix of Datta and Chakrabarty (2023), who proved consistency of the MLE in a quotient of Euclidean space; and (iii) the manifold asymptotic-efficiency framework of Sun, Lin and Liu (2025), in which asymptotic normality and regularity of estimators taking values in a Riemannian manifold are formulated through exponential maps, logarithms and parallel transport. Our object is the MLE  $\hat{\Lambda}$  of the loading matrix  $\Lambda$  in the PPCA model  $\Sigma = \Lambda\Lambda^\top + \sigma^2 I_p$ , viewed as a point of the quotient manifold  $\mathbb{R}_*^{p \times m}/O(m)$ , which we show carries the Procrustes (Bures–Wasserstein) geometry with closed-form exponential and logarithm maps. We (a) give an exact first-order expression of the quotient logarithm as a horizontal projection, showing that passing to the quotient replaces the resonant eigengap denominators  $\alpha_i - \alpha_j$  of classical eigenvector CLTs by the non-resonant  $\alpha_i + \alpha_j$ ; (b) prove a complete quotient-space CLT in the classical regime ( $p$  fixed), in the exact sense of Sun–Lin–Liu; (c) in the high-dimensional regime  $p/n \rightarrow c > 0$  prove that the MLE is *inconsistent in the quotient metric* with an explicitly computed asymptotic bias, mirroring the noise-variance bias of Passemier–Li–Yao; (d) construct a bias-corrected loading estimator  $\hat{\Lambda}_*$  by inverting the spike map  $\psi$  of Baik–Silverstein and de-biasing the eigenvector cosines; and (e) establish, modulo an imported joint eigenstructure CLT from random matrix theory (Bai–Yao 2008; Paul 2007; Bao–Ding–Wang–Wang 2022),  $\sqrt{n}$ -asymptotic normality of  $\hat{\Lambda}_*$  in the quotient tangent space, in the “visible block” and along fixed transverse directions, with fully explicit Jacobians and, in several cases, fully explicit limiting variances such as  $\varpi_i^2 = \sigma^2 a_i / (\alpha_i \psi'(\alpha_i))$  for transverse projections. All statements reduce continuously to the classical results as  $c \rightarrow 0$ . We carefully flag which ingredients are proved here, which are imported, and which remain open (notably tightness of the full  $p$ -dimensional log-map and a convolution/efficiency theorem for growing dimension).

## 1 Introduction

The PPCA model of Tipping and Bishop [4] writes an observed vector  $x \in \mathbb{R}^p$  as  $x = \Lambda f + e + \mu$  with latent factor  $f \sim N(0, I_m)$ ,  $e \sim N(0, \sigma^2 I_p)$ , so that  $x \sim N(\mu, \Sigma)$  with

$$\Sigma = \Lambda\Lambda^\top + \sigma^2 I_p, \quad \Lambda \in \mathbb{R}^{p \times m}, \text{ rank } \Lambda = m, \sigma^2 > 0. \quad (1)$$

The likelihood depends on  $\Lambda$  only through  $\Phi = \Lambda\Lambda^\top$ : the model is invariant under  $\Lambda \mapsto \Lambda R$ ,  $R \in O(m)$ , and the MLE is determined only up to this rotation [4]. Three papers motivate what follows.

(P1) *Passemier–Li–Yao* [1]. In the proportional regime  $p, n \rightarrow \infty$ ,  $c_n = p/(n-1) \rightarrow c > 0$ , the ML noise-variance estimator  $\hat{\sigma}^2 = (p-m)^{-1} \sum_{i>m} \lambda_{n,i}$  ( $\lambda_{n,i}$  the sample-covariance eigenvalues) acquires a negative bias  $-\sigma^2 \sqrt{2c} b(\sigma^2)/(p-m)$ , which they compute by random matrix theory and remove, obtaining a bias-corrected  $\hat{\sigma}_*^2$  with  $\frac{p-m}{\sigma^2 \sqrt{2c_n}} (\hat{\sigma}_*^2 - \sigma^2) \rightsquigarrow N(0, 1)$ ; in particular  $\hat{\sigma}_*^2 - \sigma^2 = O_P(n^{-1})$ , i.e. the noise variance is *super*- $\sqrt{n}$ -consistent. Their analysis rests on the spiked-model limits of Baik–Silverstein [5] and the spike CLT of Bai–Yao [6].

(P2) *Datta–Chakrabarty* [2]. The rotational symmetry makes  $(\Lambda, \sigma^2)$  non-identifiable; they pass to a quotient of the parameter space, metrized following Redner, and prove (strong) consistency of the MLE in that quotient, for fixed  $p$ . They leave rates and asymptotic distributions open (their Section 8).

(P3) *Sun–Lin–Liu* [3]. For a parameter taking values in a Riemannian manifold  $(\Psi, g)$ , asymptotic normality of an estimator  $\hat{\psi}_n$  of  $\psi$  is formulated as  $\sqrt{n} \text{Exp}_{\hat{\psi}_n}^{-1} \hat{\psi}_n \rightsquigarrow Z$  in the tangent space  $T_{\hat{\psi}_n} \Psi$ , and *regularity* as  $\Pi_{\hat{\psi}_n, h}^{\psi} \sqrt{n} \text{Exp}_{\hat{\psi}_n, h}^{-1} \hat{\psi}_n \rightsquigarrow Z$  under local perturbations  $\psi_{n, h} = \text{Exp}_{\psi}(h/\sqrt{n})$ , where  $\Pi$  is parallel transport.

The present notes carry out the program suggested by combining the three: formulate the loading-matrix problem on the quotient manifold, define asymptotic normality à la (P3), and use the random-matrix toolkit of (P1) to obtain the high-dimensional limit theory that (P2) left open. Because the sample eigenvectors of  $S_n$  are *inconsistent* when  $c > 0$ , the story necessarily splits into a bias identification/correction step (exactly parallel to (P1)’s treatment of  $\hat{\sigma}^2$ ) and a CLT step, and because  $p = p_n \rightarrow \infty$  the CLT can only be stated for suitable finite-dimensional functionals of the quotient logarithm. We are explicit below about the status of each claim: **[proved]** (self-contained argument sketched in the appendix), **[imported]** (a result quoted from the literature), or **[conjectural]** (heuristic, requiring further work).

Two remarks on strategy, following the two attack routes proposed for this problem. First, since the Gaussian likelihood (1) is an exponential family in  $\Sigma^{-1}$ , the constrained MLE of  $\Sigma$  coincides with the method-of-moments fit obtained from the top- $m$  eigenpairs of  $S_n$  and the average of the remaining eigenvalues; hence CLTs for the sample eigenstructure *are* CLTs for the MLE, and covariance-level results transfer to the loading matrix through the isometry  $[\Lambda] \leftrightarrow \Phi = \Lambda \Lambda^\top$  described in Section 2. Second, the direct random-matrix route acts on the eigenpairs themselves; this is the route we quantify. The two routes agree because the quotient geometry of loadings *is* the Bures–Wasserstein geometry of the covariance factor  $\Phi$ .

## 2 The quotient geometry of loading matrices

Let  $\mathbb{R}_*^{p \times m}$  be the full-column-rank matrices;  $O(m)$  acts freely and isometrically on the right on the flat space  $(\mathbb{R}_*^{p \times m}, \langle \cdot, \cdot \rangle_F)$ . The orbit space

$$\mathcal{M}_{p, m} := \mathbb{R}_*^{p \times m} / O(m), \quad [W] := \{WR : R \in O(m)\},$$

is a smooth quotient manifold (the action is free and proper), and the map  $[W] \mapsto \Phi = WW^\top$  identifies it with the manifold  $S_m^+(p)$  of rank- $m$  positive-semidefinite  $p \times p$  matrices. The quotient (orbit) metric is the *Procrustes distance*

$$\bar{d}([W_1], [W_2]) = \min_{R \in O(m)} \|W_1 R - W_2\|_F = \left( \text{tr } \Phi_1 + \text{tr } \Phi_2 - 2 \sum_{i=1}^m s_i(W_1^\top W_2) \right)^{1/2}, \quad (2)$$

with  $s_i$  the singular values; under the identification this is exactly the Bures–Wasserstein distance between  $\Phi_1$  and  $\Phi_2$  [10, 11]. The vertical and horizontal spaces at  $W$  are

$$V_W = \{W\Omega : \Omega^\top = -\Omega\}, \quad H_W = \{Z \in \mathbb{R}^{p \times m} : W^\top Z = Z^\top W\}, \quad (3)$$

i.e.  $Z$  is horizontal iff  $W^\top Z$  is symmetric. Because the total space is flat and the action isometric, quotient geodesics lift to horizontal straight lines [11]: for  $\xi \in H_W$ ,

$$\text{Exp}_{[W]}(\xi) = [W + \xi], \quad \text{Log}_{[W_0]}([W]) = WR^* - W_0, \quad R^* = \arg \min_{R \in O(m)} \|WR - W_0\|_F, \quad (4)$$

whenever  $W^\top W_0$  is nonsingular; then  $R^* = PQ^\top$  is the orthogonal polar factor of  $W^\top W_0 = PSQ^\top$  (SVD), the logarithm is horizontal at  $W_0$ , unique, and  $\|\text{Log}_{[W_0]}([W])\|_F = \bar{d}([W], [W_0])$ .

**Lemma 2.1** (The Redner-type quotient pseudometric is the orbit distance). *Let  $d_1(W, W') = \inf\{\sum_k d(x_k, y_k)\}$  be the chaining pseudometric of [2] (their Section 4.1) associated with the equivalence  $W \sim WR$ ,  $R \in O(m)$ . Since orbits of the compact group  $O(m)$  are compact and the action is isometric, the chaining infimum is attained in a single step and  $d_1(W, W') = \min_R \|WR - W'\|_F = \bar{d}([W], [W'])$ . Consequently the metric quotient used in [2] coincides, on loading matrices, with the Riemannian quotient above, and the pathologies of general quotients (their Appendix 11.1) do not occur. [proved]*

Fix now a representative of the truth. Under the standard identification constraints ([1], Section 1)  $\Gamma = \Lambda^\top \Lambda = \Delta = \text{diag}(\alpha_1 > \dots > \alpha_m > 0)$ , so we may write

$$\Lambda_0 = U_0 \Delta^{1/2}, \quad U_0^\top U_0 = I_m, \quad \Sigma_0 = \Lambda_0 \Lambda_0^\top + \sigma^2 I_p, \quad (5)$$

whose spectrum is  $(\alpha_1 + \sigma^2, \dots, \alpha_m + \sigma^2, \sigma^2, \dots, \sigma^2)$ ; write  $a_i := \alpha_i + \sigma^2$  and  $u_1, \dots, u_m$  for the columns of  $U_0$ .

**First-order structure of the logarithm.** For  $D \in \mathbb{R}^{p \times m}$  define the *horizontal projector* at  $\Lambda_0$ ,

$$\mathbf{P}^H(D) = D + \Lambda_0 A(D), \quad A(D)_{ji} = -\frac{2[\text{skew}(\Lambda_0^\top D)]_{ji}}{\alpha_i + \alpha_j}, \quad (6)$$

where  $\text{skew}(M) = (M - M^\top)/2$ ;  $A(D)$  is the unique skew-symmetric solution of the Sylvester equation  $\frac{1}{2}(\Delta A + A\Delta) = -\text{skew}(\Lambda_0^\top D)$ , and  $\mathbf{P}^H$  is the orthogonal projection of  $\mathbb{R}^{p \times m}$  onto  $H_{\Lambda_0}$  along  $V_{\Lambda_0}$ .

**Lemma 2.2** (First-order expansion of the quotient logarithm). *Let  $W_n \in \mathbb{R}_*^{p \times m}$  (possibly with  $p = p_n$  growing) admit rotations  $\tilde{R}_n \in O(m)$  with  $D_n := W_n \tilde{R}_n - \Lambda_0 \rightarrow 0$  in Frobenius norm and  $\liminf s_m(W_n^\top \Lambda_0) > 0$ . Then*

$$\text{Log}_{[\Lambda_0]}([W_n]) = \mathbf{P}^H(D_n) + O(\|D_n\|_F^2),$$

and  $\bar{d}([W_n], [\Lambda_0]) = \|\mathbf{P}^H(D_n)\|_F + O(\|D_n\|_F^2)$ . [proved] (Appendix A.1.)

The vertical correction  $\Lambda_0 A(D)$  has columns in  $\text{span}(U_0)$ : quotienting only re-mixes the “visible”  $m \times m$  block. Reading (6) in the block coordinates below exhibits the key regularization phenomenon: an antisymmetric perturbation of the visible block, which for eigenvectors carries the resonant factor  $(\alpha_i - \alpha_j)^{-1}$ , is a pure gauge (vertical) direction, and after projection only the *symmetric* part, weighted by  $(\alpha_i + \alpha_j)^{-1}$ , survives (Theorem 3.1).

**Coordinates.** For a  $p \times m$  tangent object  $\xi$  we use the coordinates  $\xi^{(ji)} := u_j^\top \xi e_i$  ( $i, j \leq m$ ; the *visible block*  $U_0^\top \xi$ ) and  $\xi^{(vi)} := v^\top \xi e_i$  for unit  $v \perp \text{span}(U_0)$  (*transverse coordinates*).

### Asymptotic normality in the quotient space, à la [3].

**Definition 2.3** (Quotient asymptotic normality; fixed  $p$ ). An estimator sequence  $\hat{\Lambda}_n$  is *asymptotically normal at  $[\Lambda_0]$  in the quotient space* if  $\sqrt{n} \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_n]) \rightsquigarrow \mathcal{Z}$ , a centered Gaussian element of the horizontal space  $H_{\Lambda_0} \cong T_{[\Lambda_0]}\mathcal{M}_{p,m}$ . It is *regular at  $[\Lambda_0]$*  if for every horizontal  $h$ , with  $[\Lambda_{n,h}] := \text{Exp}_{[\Lambda_0]}(h/\sqrt{n})$  and data generated from  $\Lambda_{n,h}$ ,

$$\Pi_{[\Lambda_{n,h}]}^{[\Lambda_0]} \sqrt{n} \text{Log}_{[\Lambda_{n,h}]}([\hat{\Lambda}_n]) \rightsquigarrow \mathcal{Z} \quad \text{with the same } \mathcal{Z},$$

$\Pi$  denoting parallel transport along the connecting geodesic (Definition 3.5 of [3]).

**Definition 2.4** (Anchored/cylindrical version;  $p = p_n \rightarrow \infty$ ). When  $p_n \rightarrow \infty$  the manifolds change with  $n$  and weak convergence of the full logarithm in a fixed space is unavailable. We say  $\hat{\Lambda}_n$  is *cylindrically asymptotically normal at  $[\Lambda_0]$*  (relative to an anchor sequence  $U_0 = U_0^{(n)}$ ,  $v = v^{(n)} \perp U_0$ ) if every fixed finite collection of visible and transverse coordinates of  $\sqrt{n} \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_n])$  converges jointly to a centered Gaussian vector. Regularity is required coordinatewise in the same sense.

*Remark 2.5.* Definition 2.4 is the natural triangular-array weakening of [3]: their framework presumes a fixed manifold; here the parameter manifold  $\mathcal{M}_{p_n,m}$  itself grows. Establishing a genuinely infinite-dimensional (tight) version is open; see Section 6.

### 3 The MLE and the classical quotient CLT ( $p$ fixed)

Let  $S_n = (n-1)^{-1} \sum_i (x_i - \bar{x})(x_i - \bar{x})^\top$  with eigenvalues  $\lambda_{n,1} \geq \dots \geq \lambda_{n,p}$  and orthonormal eigenvectors  $\hat{u}_1, \dots, \hat{u}_p$ , signs fixed by  $u_i^\top \hat{u}_i \geq 0$ . The MLE of [4] is

$$\hat{\Lambda} = \hat{U}_m (\hat{\Delta}_m - \hat{\sigma}^2 I_m)^{1/2} R, \quad \hat{\sigma}^2 = \frac{1}{p-m} \sum_{i=m+1}^p \lambda_{n,i}, \quad (7)$$

$\hat{U}_m = [\hat{u}_1, \dots, \hat{u}_m]$ ,  $\hat{\Delta}_m = \text{diag}(\lambda_{n,1}, \dots, \lambda_{n,m})$ ,  $R \in O(m)$  arbitrary: precisely an equivalence class  $[\hat{\Lambda}] \in \mathcal{M}_{p,m}$ . It is also the method-of-moments fit of  $\Sigma$  under (1), so all results below are simultaneously statements about the constrained covariance MLE  $\hat{\Sigma} = \hat{\Lambda} \hat{\Lambda}^\top + \hat{\sigma}^2 I_p$  in Bures–Wasserstein geometry.

In the classical regime the eigenstructure CLT (Anderson [12]; Gaussian data, distinct  $a_1 > \dots > a_m > \sigma^2$ ) states that jointly, with  $E_n = \sqrt{n}(S_n - \Sigma_0)$ ,

$$h_i := \lim u_i^\top E_n u_i \sim N(0, 2a_i^2), \quad g_{ij} := \lim u_j^\top E_n u_i \sim N(0, a_i a_j) \quad (i \neq j \leq m),$$

$$s_0 := \lim \sqrt{n}(\hat{\sigma}^2 - \sigma^2) \sim N\left(0, \frac{2\sigma^4}{p-m}\right), \quad \zeta_i := \lim \frac{\sqrt{n}}{\sqrt{a_i \sigma^2}} (I - U_0 U_0^\top) E_n u_i \sim N(0, I_{p-m}),$$

all mutually independent, and  $\sqrt{n}(\hat{u}_i - u_i) \rightarrow \sum_{j \neq i, j \leq m} \frac{g_{ij}}{a_i - a_j} u_j + \frac{\sqrt{a_i} \sigma}{\alpha_i} \zeta_i$ ,  $\sqrt{n}(\lambda_{n,i} - a_i) \rightarrow h_i$ .

**Theorem 3.1** (Classical quotient CLT). *Fix  $p$ , let the data be Gaussian and  $\alpha_1 > \dots > \alpha_m > 0$  be distinct. Then  $[\hat{\Lambda}]$  is asymptotically normal and regular at  $[\Lambda_0]$  in the sense of Definition 2.3:*

$\sqrt{n} \text{Log}_{[\Lambda_0]}([\hat{\Lambda}]) \rightsquigarrow \mathcal{Z} \in H_{\Lambda_0}$ , a centered Gaussian with independent components

$$\mathcal{Z}^{(ii)} = \frac{h_i - s_0}{2\sqrt{\alpha_i}} \quad \Rightarrow \quad \text{Var } \mathcal{Z}^{(ii)} = \frac{a_i^2}{2\alpha_i} + \frac{\sigma^4}{2\alpha_i(p-m)}, \quad (8)$$

$$(\mathcal{Z}^{(ji)}, \mathcal{Z}^{(ij)}) = \frac{(\sqrt{\alpha_i}, \sqrt{\alpha_j})}{\alpha_i + \alpha_j} g_{ij} \quad \Rightarrow \quad \text{Var } \mathcal{Z}^{(ji)} = \frac{\alpha_i a_i a_j}{(\alpha_i + \alpha_j)^2} \quad (i \neq j \leq m), \quad (9)$$

$$\mathcal{Z}^{(vi)} = \sigma \sqrt{\frac{a_i}{\alpha_i}} \langle v, \zeta_i \rangle \quad \Rightarrow \quad \text{Var } \mathcal{Z}^{(vi)} = \frac{\sigma^2 a_i}{\alpha_i} \quad (v \perp U_0), \quad (10)$$

the only dependence across components being the common  $s_0$  in (8), with  $\text{Cov}(\mathcal{Z}^{(ii)}, \mathcal{Z}^{(jj)}) = \sigma^4 / (2\sqrt{\alpha_i \alpha_j} (p-m))$ ; and  $\Lambda_0^\top \mathcal{Z}$  is symmetric a.s. (the limit is horizontal). **[proved]** (Appendix A.2.)

*Remark 3.2* (Gauge absorption of the eigengap). In (9) the eigenvector resonance  $(\alpha_i - \alpha_j)^{-1}$  has disappeared: the raw visible-block perturbations are  $V^{(ji)} = \sqrt{\alpha_i} g_{ij} / (\alpha_i - \alpha_j)$ , but their antisymmetric part is vertical, and the horizontal projection (6) yields

$$\mathbf{P}^H(V)^{(ji)} = \frac{\alpha_i V^{(ji)} + \sqrt{\alpha_i \alpha_j} V^{(ij)}}{\alpha_i + \alpha_j} = \frac{\sqrt{\alpha_i} g_{ij}}{\alpha_i + \alpha_j}.$$

Thus the quotient CLT remains nondegenerate and finite-variance as  $\alpha_i \rightarrow \alpha_j$ : the ill-conditioning of principal *axes* under near-degeneracy is an artifact of the non-identifiable parametrization, not of the identifiable parameter  $[\Lambda]$  (equivalently  $\Phi$ ). This is the exact loading-matrix analogue of a phenomenon the quotient viewpoint of [2] is designed to capture, made quantitative.

*Remark 3.3* (Regularity and parallel transport). For  $h \in H_{\Lambda_0}$  and  $\Lambda_{n,h} = \Lambda_0 + h/\sqrt{n}$  the model is a smooth parametric family; local asymptotic normality plus Le Cam's third lemma give the  $h$ -free limit as in [3], and since  $\bar{d}([\Lambda_{n,h}], [\Lambda_0]) = O(n^{-1/2})$ , parallel transport differs from the coordinate identification by  $O(n^{-1})$  (curvature  $\times$  distance<sup>2</sup>), which vanishes after  $\sqrt{n}$ -scaling (Lemma 5.1). The MLE is therefore *regular*; a Hájek–Le Cam convolution/efficiency statement in the quotient then follows from Theorem 3.1 of [3] once the DQM score calculus is written in horizontal coordinates – we record this as **[imported + routine]** and focus on the high-dimensional case.

## 4 High-dimensional regime $p/n \rightarrow c > 0$

Assume throughout: Gaussian data;  $m$  fixed;  $p, n \rightarrow \infty$  with  $c_n = p/(n-1) \rightarrow c \in (0, \infty)$ ; distinct supercritical spikes

$$\alpha_1 > \cdots > \alpha_m > \sqrt{c} \sigma^2 \quad (\text{equivalently } \bar{\alpha}_i := \alpha_i / \sigma^2 > \sqrt{c}). \quad (11)$$

Define, as in [1, 5],

$$\psi(\alpha) = \alpha + \sigma^2 + c\sigma^2 \left(1 + \frac{\sigma^2}{\alpha}\right), \quad \psi'(\alpha) = 1 - \frac{c\sigma^4}{\alpha^2}, \quad \rho_i := \psi(\alpha_i), \quad \kappa_i^2 := \rho_i - \sigma^2, \quad (12)$$

and the squared limiting cosine (Paul [7])

$$\theta_i = \frac{\alpha_i^2 - c\sigma^4}{\alpha_i(\alpha_i + c\sigma^2)} \in (0, 1), \quad 1 - \theta_i = \frac{c\sigma^2(\alpha_i + \sigma^2)}{\alpha_i(\alpha_i + c\sigma^2)}. \quad (13)$$

Then a.s.  $\lambda_{n,i} \rightarrow \rho_i$ ,  $\hat{\sigma}^2 \rightarrow \sigma^2$  (indeed  $\hat{\sigma}^2 - \sigma^2 = O_P(n^{-1})$  by Theorem 1 of [1]),  $u_i^\top \hat{u}_i \rightarrow \sqrt{\theta_i}$ ,  $u_j^\top \hat{u}_i \rightarrow 0$  ( $j \neq i$ ), and the residual mass  $1 - \theta_i$  of  $\hat{u}_i$  is delocalized over  $\text{span}(U_0)^\perp$  [5, 7]. Note  $\theta_i > 0 \iff \psi'(\alpha_i) > 0 \iff (11)$ : sub-critical directions are invisible, so (11) is also necessary for any of what follows.

#### 4.1 Inconsistency of the MLE in the quotient metric, and the exact bias

**Theorem 4.1** (Quotient-space bias of the MLE). *Under the above,*

$$\bar{d}^2([\hat{\Lambda}], [\Lambda_0]) \xrightarrow{\text{a.s.}} B(c) := \sum_{i=1}^m \left( \alpha_i + \kappa_i^2 - 2\kappa_i \sqrt{\theta_i \alpha_i} \right) = \sum_{i=1}^m \left[ (\kappa_i \sqrt{\theta_i} - \sqrt{\alpha_i})^2 + \kappa_i^2 (1 - \theta_i) \right] > 0.$$

More precisely, conditionally on the (delocalized, asymptotically uniform) unit vectors  $v_i \in \text{span}(U_0)^\perp$  carrying the transverse mass of  $\hat{u}_i$ ,

$$\text{Log}_{[\Lambda_0]}([\hat{\Lambda}]) \longrightarrow \sum_{i=1}^m \left[ (\kappa_i \sqrt{\theta_i} - \sqrt{\alpha_i}) u_i + \kappa_i \sqrt{1 - \theta_i} v_i \right] e_i^\top \quad \text{a.s.}$$

$B(c) \rightarrow 0$  as  $c \rightarrow 0$  and  $B$  is increasing in  $c$  near 0: the MLE of the loading matrix is inconsistent in the quotient space for every  $c > 0$ , with a radial (length) bias  $\kappa_i \sqrt{\theta_i} - \sqrt{\alpha_i}$  and a transverse (angle) bias of squared size  $\kappa_i^2 (1 - \theta_i)$ . **[proved]** (Appendix A.3; the ingredients are (12), (13) and continuity of the Procrustes alignment, whose limit is  $R^* \rightarrow I_m$  under our sign convention.)

This is the loading-matrix counterpart of the negative bias of  $\hat{\sigma}^2$  identified in [1], with two differences: the bias here is  $O(1)$  rather than  $O(n^{-1})$  (eigenvector inconsistency), and it has both a deterministic radial part and a random-direction transverse part. As in [1], the remedy is to *invert the asymptotic map*.

#### 4.2 Bias-corrected loading estimator

Solving  $\lambda = \psi(\alpha)$  for  $\alpha$  gives, with  $\hat{\sigma}_\star^2$  the bias-corrected noise variance of [1] (their eq. (7); the raw  $\hat{\sigma}^2$  may also be used since both are  $\sigma^2 + O_P(n^{-1})$ ),

$$\hat{\alpha}_i = \frac{1}{2} \left[ \lambda_{n,i} - \hat{\sigma}_\star^2 (1 + c_n) + \sqrt{(\lambda_{n,i} - \hat{\sigma}_\star^2 (1 + c_n))^2 - 4c_n \hat{\sigma}_\star^4} \right], \quad (14)$$

the discriminant being nonnegative exactly when  $\lambda_{n,i} \geq \hat{\sigma}_\star^2 (1 + \sqrt{c_n})^2$ , i.e. when the  $i$ th eigenvalue exceeds the (estimated) Baik–Ben Arous–Péché bulk edge – the estimator is defined precisely on the detectable event. This is the spike estimator used for the real-data analysis of [1] (their Section 2). Set  $\hat{\theta}_i = \theta(\hat{\alpha}_i; \hat{\sigma}_\star^2, c_n)$  by (13), and define the *de-biased loading estimator*

$$\hat{\Lambda}_\star := \hat{U}_m \text{diag}(\hat{\beta}_1, \dots, \hat{\beta}_m), \quad \hat{\beta}_i := (\hat{\alpha}_i / \hat{\theta}_i)^{1/2}, \quad \beta(\alpha) := \sqrt{\alpha / \theta(\alpha)} = \alpha \sqrt{\frac{\alpha + c\sigma^2}{\alpha^2 - c\sigma^4}}. \quad (15)$$

By construction  $U_0^\top \hat{\Lambda}_\star \rightarrow \text{diag}(\sqrt{\theta_i}) \text{diag}(\beta_i) = \Delta^{1/2}$ : the *visible block* of the bias is removed. The transverse component cannot be removed by any rotation-equivariant procedure – eigenvector inconsistency is information-theoretic in this regime – and indeed  $\bar{d}^2([\hat{\Lambda}_\star], [\Lambda_0]) \rightarrow B_\star(c) = \sum_i \alpha_i (1 - \theta_i) / \theta_i > 0$ . What is available, and is the content of the next subsection, is  $\sqrt{n}$ -normality of all anchored coordinates of the logarithm: the visible block fluctuates around 0 at rate  $n^{-1/2}$ , and each fixed transverse projection of the  $O(1)$  delocalized component is itself  $O_P(n^{-1/2})$  and asymptotically normal.

#### 4.3 Imported random-matrix CLTs

Let  $P_n := U_0^\top \hat{U}_m \in \mathbb{R}^{m \times m}$  (sign convention: diagonal  $\geq 0$ ) and  $\Theta := \text{diag}(\theta_1, \dots, \theta_m)$ .

**Assumption 4.2** (Joint visible-block eigenstructure CLT). Under (11) with Gaussian data, jointly,

$$T_n := \left( \sqrt{n}(\lambda_{n,i} - \rho_i)_{i \leq m}, \sqrt{n}(P_n - \Theta^{1/2}) \right) \rightsquigarrow (h, \mathcal{P}),$$

a centered Gaussian vector with covariance  $\Xi$ , where the eigenvalue block is explicit (Bai–Yao [6], real Gaussian case, simple spikes):

$$h_i \text{ independent, } \quad \text{Var}(h_i) = \varsigma_i^2 = 2a_i^2 \psi'(\alpha_i) = 2a_i^2 \left( 1 - \frac{c\sigma^4}{\alpha_i^2} \right). \quad (16)$$

Moreover  $\hat{\sigma}_*^2 - \sigma^2 = O_P(n^{-1})$  (Theorem 2 of [1]), hence negligible at scale  $\sqrt{n}$ .

*Status.* **[imported]**: (16) is Bai–Yao [6]; the diagonal entries of  $\mathcal{P}$  (fluctuations of the cosines  $u_i^\top \hat{u}_i$  around  $\sqrt{\theta_i}$ ) are covered by Paul [7] (his Theorem 5, via the block decomposition  $\hat{u}_i = (\hat{u}_{A,i}, \hat{u}_{B,i})$ ), and the joint law of all entries of  $(h, \mathcal{P})$ , including off-diagonals  $\sqrt{n} u_j^\top \hat{u}_i$ , is established by Bao–Ding–Wang–Wang [8] in the supercritical regime. What we use below is only the existence of the joint Gaussian limit and (16); we denote the remaining blocks of  $\Xi$  symbolically ( $\omega_i^2 := \text{Var } \mathcal{P}_{ii}$ ,  $\tau_{ij}^2 := \text{Var } \mathcal{P}_{ji}$ , correlations  $r_i := \text{Corr}(\mathcal{P}_{ii}, h_i)$ , etc.). Their classical limits are  $\omega_i^2 \rightarrow 0$ ,  $\tau_{ij}^2 \rightarrow a_i a_j / (\alpha_i - \alpha_j)^2$ , consistent with Section 3. Transcribing the exact finite- $c$  expressions of [8] into the present normalization is bookkeeping we have not completed; every statement below is exact conditional on  $\Xi$ .

**Assumption 4.3** (Transverse projections). For any fixed sequence of unit vectors  $v = v^{(n)} \perp U_0$ , jointly with Assumption 4.2,  $\sqrt{n} v^\top \hat{u}_i \rightsquigarrow N(0, \nu_i^2)$  with

$$\nu_i^2 = \frac{1 - \theta_i}{c} = \frac{\sigma^2(\alpha_i + \sigma^2)}{\alpha_i(\alpha_i + c\sigma^2)}, \quad (17)$$

independent across orthonormal transverse directions and across  $i$  at first order. **[imported/derived]**: the delocalization  $\|(I - U_0 U_0^\top) \hat{u}_i\|^2 \rightarrow 1 - \theta_i$  spread over  $p - m \sim cn$  dimensions gives (17) by isotropy; Gaussianity of such projections follows from isotropic local laws / eigenvector delocalization CLTs ([9, 8]). The value (17) is forced by the consistency check below, but we have not re-derived the rigorous statement in this exact form: treat (17) as **[conjectural constant, established phenomenon]**.

#### 4.4 Main results: quotient asymptotic normality of $\hat{\Lambda}_*$

Define the visible-block symmetrizer  $\mathcal{H}_\Delta : \mathbb{R}^{m \times m} \rightarrow \mathbb{R}^{m \times m}$ ,

$$\mathcal{H}_\Delta(M) = M - \Delta^{1/2} \text{Sylv}_\Delta(\text{skew}(\Delta^{1/2} M)), \quad \text{Sylv}_\Delta(K)_{ji} = \frac{2K_{ji}}{\alpha_i + \alpha_j}, \quad (18)$$

i.e. the visible-block action of  $P^H$ ; explicitly  $\mathcal{H}_\Delta(M)_{ji} = (\alpha_i M_{ji} + \sqrt{\alpha_i \alpha_j} M_{ij}) / (\alpha_i + \alpha_j)$  for  $j \neq i$  and  $\mathcal{H}_\Delta(M)_{ii} = M_{ii}$ .

**Theorem 4.4** (Visible-block quotient CLT,  $c > 0$ ). *Let (11) and Assumption 4.2 hold and let  $\hat{\Lambda}_*$  be (15). Then*

$$\sqrt{n} U_0^\top \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_*]) \rightsquigarrow \mathcal{Y} = \mathcal{H}_\Delta \left( \mathcal{P} \text{diag}(\beta_i) + \Theta^{1/2} \text{diag} \left( \frac{\beta'(\alpha_i)}{\psi'(\alpha_i)} h_i \right) \right), \quad (19)$$

a centered Gaussian on  $\{M : \Delta^{1/2}M \text{ symmetric}\} \cong T_{[\Lambda_0]} \mathcal{M}_{p,m}|_{\text{visible}}$ , with  $\beta$  as in (15) and

$$(\beta^2)'(\alpha) = \frac{\alpha(\alpha + \sqrt{c}\sigma^2)^2(\alpha - 2\sqrt{c}\sigma^2)}{(\alpha^2 - c\sigma^4)^2}, \quad \beta' = \frac{(\beta^2)'}{2\beta}. \quad (20)$$

In particular:

(i) Diagonal coordinates:  $\mathcal{Y}_{ii} = \beta_i \mathcal{P}_{ii} + \sqrt{\theta_i} \frac{\beta'(\alpha_i)}{\psi'(\alpha_i)} h_i$ , so

$$\text{Var } \mathcal{Y}_{ii} = \beta_i^2 \omega_i^2 + \theta_i \frac{\beta'(\alpha_i)^2}{\psi'(\alpha_i)^2} \varsigma_i^2 + 2\beta_i \sqrt{\theta_i} \frac{\beta'(\alpha_i)}{\psi'(\alpha_i)} \varsigma_i \omega_i r_i, \quad \theta_i \frac{\beta'^2}{\psi'^2} \varsigma_i^2 = \frac{2a_i^2 \theta_i \beta'(\alpha_i)^2}{\psi'(\alpha_i)}.$$

(ii) Off-diagonal pairs ( $i \neq j$ ): with  $M_{ji} = \beta_i \mathcal{P}_{ji}$ ,  $\mathcal{Y}_{ji} = (\alpha_i \beta_i \mathcal{P}_{ji} + \sqrt{\alpha_i \alpha_j} \beta_j \mathcal{P}_{ij})/(\alpha_i + \alpha_j)$ : the eigengap-resonant statistics  $\mathcal{P}_{ji}$  enter only through the  $(\alpha_i + \alpha_j)$ -weighted symmetric combination, extending Remark 3.2 to  $c > 0$ .

(iii) As  $c \rightarrow 0$ :  $\beta_i \rightarrow \sqrt{\alpha_i}$ ,  $\beta' \rightarrow 1/(2\sqrt{\alpha_i})$ ,  $\psi' \rightarrow 1$ ,  $\theta_i \rightarrow 1$ ,  $\varsigma_i^2 \rightarrow 2a_i^2$ ,  $\omega_i \rightarrow 0$ , and (19) reduces exactly to the visible part of Theorem 3.1 (e.g.  $\text{Var } \mathcal{Y}_{ii} \rightarrow a_i^2/(2\alpha_i)$ ).

**[proved modulo Assumption 4.2]** (Appendix A.4: the proof is a delta method on the quotient combining Lemma 2.2, the chain rule through  $\psi^{-1}$  and  $\beta$ , and Slutsky for  $\hat{\sigma}_\star^2$  via [1].)

**Theorem 4.5** (Transverse quotient CLT). Under (11) and Assumptions 4.2–4.3, for any fixed unit  $v = v^{(n)} \perp U_0$  and each  $i \leq m$ ,

$$\sqrt{n} v^\top \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_\star]) e_i \rightsquigarrow N(0, \varpi_i^2), \quad \varpi_i^2 = \beta_i^2 \nu_i^2 = \frac{\sigma^2 a_i}{\alpha_i \psi'(\alpha_i)} = \frac{\sigma^2 \alpha_i a_i}{\alpha_i^2 - c\sigma^4}, \quad (21)$$

jointly with (19) and asymptotically independent of it, and independently across orthonormal transverse  $v$ 's. As  $c \rightarrow 0$ ,  $\varpi_i^2 \rightarrow \sigma^2 a_i / \alpha_i$ , matching (10) exactly; as  $\alpha_i \downarrow \sqrt{c}\sigma^2$ ,  $\varpi_i^2 \uparrow \infty$ , reflecting the BBP phase transition. **[proved modulo Assumption 4.3]**. Together, Theorems 4.4 and 4.5 say that  $\hat{\Lambda}_\star$  is cylindrically asymptotically normal at  $[\Lambda_0]$  in the sense of Definition 2.4.

**Corollary 4.6** (Rank one). For  $m = 1$ ,  $\mathcal{H}_\Delta = \text{id}$  and, writing  $\alpha = \alpha_1$  etc.,

$$\sqrt{n} \left( u_1^\top \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_\star]), v^\top \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_\star]) \right) \rightsquigarrow N \left( 0, \text{diag} \left( \beta^2 \omega^2 + \frac{2a^2 \theta \beta'^2}{\psi'} + 2\beta \sqrt{\theta} \frac{\beta'}{\psi'} \varsigma \omega r, \frac{\sigma^2 \alpha a}{\alpha^2 - c\sigma^4} \right) \right).$$

*Remark 4.7* (Which bias correction, and why it matters). Without the correction (15) – i.e. for the raw MLE (7) – the visible block of the logarithm is centered at the nonzero constant  $\text{diag}(\kappa_i \sqrt{\theta_i} - \sqrt{\alpha_i})$  of Theorem 4.1, so no centered CLT at the truth is possible: exactly as in [1], the high-dimensional theory forces the bias correction, and the plug-in inversion (14) is the loading-space analogue of their  $\hat{\sigma}_\star^2$  (indeed it *uses* their  $\hat{\sigma}_\star^2$ ). One may alternatively center at the biased limit: Theorem 4.4 holds verbatim for  $\hat{\Lambda}$  with  $\beta$  replaced by  $\kappa\sqrt{\theta}$ -type constants and the limit taken around the deterministic biased point; the corrected form is the statistically meaningful one.

*Remark 4.8* (Non-Gaussian data). Theorem 3 of [1] extends the noise-variance CLT to  $x_i = \Sigma^{1/2} y_i$  with i.i.d. standardized entries and fourth-moment parameter  $\gamma$ ; correspondingly, in Assumption 4.2 the eigenvalue variances become  $\gamma$ -dependent (Bai–Yao [6] covers this) and the bootstrap estimator  $\hat{\gamma}^*$  of [1] can be plugged into  $\hat{\sigma}_{\star\star}^2$  and hence into (14)–(15). The structure of (19)–(21) is unchanged; only  $\Xi$  changes. **[routine given the above]**

## 5 Regularity in the sense of Sun–Lin–Liu

**Lemma 5.1** (Transport is negligible at CLT scale). *Let  $[\Lambda_{n,h}] = \text{Exp}_{[\Lambda_0]}(h/\sqrt{n})$  for a fixed horizontal  $h$  (with  $U_0$ -anchored profile). For any tangent vectors  $\xi_n$  with  $\|\xi_n\|_F = O_P(1)$ ,*

$$\Pi_{[\Lambda_{n,h}]}^{[\Lambda_0]} \xi_n = \iota_n(\xi_n) + O_P(n^{-1}),$$

where  $\iota_n$  is the coordinate identification of the two horizontal spaces through the common ambient  $\mathbb{R}^{p \times m}$  (align by the connecting Procrustes rotation). Consequently, in Definitions 2.3–2.4 the parallel transport may be replaced by  $\iota_n$  without changing any limit. **[proved]** (Appendix A.6: quotient curvature is bounded on compacta of the regular stratum; transport along a geodesic of length  $O(n^{-1/2})$  distorts vectors by  $O(\text{curv} \times \text{length}^2)$ .)

**Proposition 5.2** (Regularity, visible directions). *In the setting of Theorem 4.4, let  $h$  be horizontal with visible profile ( $h = U_0 H$ ,  $\Delta^{1/2} H$  symmetric,  $H$  fixed) and generate data from  $\Lambda_{n,h}$ . Then the visible-block limit law  $\mathcal{Y}$  in (19) is unchanged:*

$$\Pi_{[\Lambda_{n,h}]}^{[\Lambda_0]} \sqrt{n} U_0^\top \text{Log}_{[\Lambda_{n,h}]}([\hat{\Lambda}_\star]) \rightsquigarrow \mathcal{Y}.$$

Status: **[proved modulo Assumption 4.2']**, where ( $A'$ ) is the (mild) strengthening that the joint CLT of Assumption 4.2 holds locally uniformly in the spike parameters – true in the cited sources, whose limits are continuous in  $(\alpha_i)$ ; the proof is then Lemma 5.1 plus a chain rule, since perturbing  $\alpha \mapsto \alpha + O(n^{-1/2})$  moves  $(\rho_i, \theta_i, \beta_i)$  by  $O(n^{-1/2})$  with derivatives exactly cancelled by the de-biasing map  $\psi^{-1}$  – the same cancellation that makes (19) centered. Transverse perturbation directions raise a genuinely new issue (they rotate  $U_0$  within a growing space) and are left open. A convolution (Hájek–Le Cam) theorem à la Theorem 3.1 of [3] for the visible block – i.e. optimality of the covariance in Theorem 4.4 among regular estimators – requires a growing-dimension LAN theory and is open; we conjecture it holds with the Fisher information computed blockwise from the spiked model.

## 6 Discussion, and what remains open

**What is established here.** (1) The identification of the quotient geometry of PPCA loadings with Procrustes/Bures–Wasserstein geometry, with closed-form Exp, Log, horizontal projection (6), and the observation (Lemma 2.1) that the abstract quotient metric of [2] is exactly this geometry. (2) A complete and fully explicit classical quotient CLT (Theorem 3.1), which appears to be new in this coordinate-free form, and the gauge-absorption phenomenon  $(\alpha_i - \alpha_j) \rightsquigarrow (\alpha_i + \alpha_j)$  (Remark 3.2). (3) The exact  $O(1)$  quotient bias of the MLE for  $c > 0$  (Theorem 4.1) and its correction (15), completing the program of [1] from  $\sigma^2$  to  $\Lambda$ . (4) Conditional on the imported eigenstructure CLTs, the cylindrical quotient CLT (Theorems 4.4–4.5) with explicit Jacobians and the closed forms (20), (21), all reducing continuously to the classical case as  $c \rightarrow 0$  – the high-dimensional counterpart, in the sense of [3], of the asymptotic distribution left open in [2].

**What is imported.** The joint Gaussian limit of the visible eigenstructure (Assumption 4.2: [6, 7, 8]) and the transverse projection CLT (Assumption 4.3: [9, 8]). The remaining bookkeeping is to transcribe the covariance  $\Xi$  of [8] into the present normalization, yielding closed forms for  $\omega_i^2, \tau_{ij}^2, r_i$  in Theorem 4.4(i)–(ii).

**What is open.** (a) *Tightness of the full logarithm:* the transverse part of  $\sqrt{n} \text{Log}_{[\Lambda_0]}([\hat{\Lambda}_\star])$  has total Frobenius mass of order  $\sqrt{n} \cdot O(1)$ ; no fixed-space weak limit exists, and Definition 2.4

(cylindrical convergence) is likely the best possible formulation without renormalizing the geometry – an interesting alternative is to rescale the transverse metric by  $p^{-1/2}$ , i.e. to work on a sequence of manifolds with collapsing fibers, and seek a functional limit. (b) *Efficiency*: a convolution theorem in growing dimension (Proposition 5.2 and beyond), requiring LAN for triangular arrays of spiked models – the natural marriage of [3] with random-matrix contiguity theory (Onatski–Moreira–Hallin). (c) *Uniformity near the phase transition*  $\alpha_m \downarrow \sqrt{c}\sigma^2$ , where  $\psi' \rightarrow 0$  and all our variances blow up; the correct limit theory there is Tracy–Widom-type, outside the Gaussian framework of Definition 2.3. (d) *Degenerate spikes*: Remark 3.2 suggests the quotient CLT survives  $\alpha_i = \alpha_j$  (unlike the eigenvector CLT); making this rigorous requires replacing the anchor  $U_0$  by the eigenspace and the visible block by a Grassmannian-times-spectrum coordinate – the quotient geometry accommodates this (the stratified structure of  $S_m^+(p)$ ), the perturbation theory must be redone.

## A Proof sketches

### A.1 Lemma 2.2

Existence/uniqueness and smoothness of the Procrustes rotation near  $\Lambda_0$ :  $W \mapsto R^*(W)$  is the orthogonal polar factor of  $W^\top \Lambda_0$ , smooth where  $W^\top \Lambda_0$  is nonsingular. Write  $W = \Lambda_0 + D$  (after  $\tilde{R}_n$ ),  $R^* = I + A + O(\|D\|^2)$  with  $A$  skew. Horizontality of the logarithm (4) at  $\Lambda_0$  requires  $\Lambda_0^\top (WR^* - \Lambda_0)$  symmetric; expanding,  $\Lambda_0^\top D + \Delta A$  must be symmetric, i.e.  $\text{skew}(\Lambda_0^\top D) + \frac{1}{2}(\Delta A + A\Delta) = 0$ , which is (6). Then  $WR^* - \Lambda_0 = D + \Lambda_0 A + O(\|D\|^2) = \mathbf{P}^H(D) + O(\|D\|^2)$ . The distance statement follows since  $\|\text{Log}\| = \bar{d}$  and  $\mathbf{P}^H$  is an orthogonal projection. (All constants depend only on  $\Delta$ , not on  $p$ : the bound is uniform in growing  $p$ .)  $\square$

### A.2 Theorem 3.1

Write  $\hat{\Lambda}e_i = \hat{\kappa}_i \hat{u}_i$ ,  $\hat{\kappa}_i = (\lambda_{n,i} - \hat{\sigma}^2)^{1/2}$ . By the classical eigen-CLT and the delta method,  $\sqrt{n}(\hat{\Lambda} - \Lambda_0)e_i \rightarrow \delta_i u_i + \sum_{j \neq i, j \leq m} \frac{\sqrt{\alpha_i}}{\alpha_i - \alpha_j} g_{ij} u_j + \sigma \sqrt{a_i/\alpha_i} \zeta_i$  with  $\delta_i = (h_i - s_0)/(2\sqrt{\alpha_i})$  (using  $a_i - a_j = \alpha_i - \alpha_j$  and  $a_i - \sigma^2 = \alpha_i$ ). Apply Lemma 2.2: transverse and diagonal coordinates are unchanged by  $\mathbf{P}^H$  ( $A$  acts only inside the visible block and  $A_{ii} = 0$ ), while for  $j \neq i$ , with  $V^{(ji)} = \sqrt{\alpha_i} g_{ij}/(\alpha_i - \alpha_j)$ ,  $V^{(ij)} = -\sqrt{\alpha_j} g_{ij}/(\alpha_i - \alpha_j)$ ,

$$\mathbf{P}^H(V)^{(ji)} = \frac{\alpha_i V^{(ji)} + \sqrt{\alpha_i \alpha_j} V^{(ij)}}{\alpha_i + \alpha_j} = \frac{\sqrt{\alpha_i} g_{ij}}{(\alpha_i + \alpha_j)(\alpha_i - \alpha_j)} (\alpha_i - \alpha_j) = \frac{\sqrt{\alpha_i} g_{ij}}{\alpha_i + \alpha_j},$$

the announced cancellation. Horizontality:  $\sqrt{\alpha_j} \mathcal{Z}^{(ji)} = \sqrt{\alpha_i \alpha_j} g_{ij}/(\alpha_i + \alpha_j) = \sqrt{\alpha_i} \mathcal{Z}^{(ij)}$ . Independence and variances are read off the classical joint law. Regularity: as Remark 3.3.  $\square$

### A.3 Theorem 4.1

By [5] and [7], a.s.  $\lambda_{n,i} \rightarrow \rho_i$ ,  $\hat{\sigma}^2 \rightarrow \sigma^2$ ,  $P_n \rightarrow \Theta^{1/2}$ , and  $\hat{u}_i = \sqrt{\theta_i} u_i + \sqrt{1 - \theta_i} v_i + o(1)$  with  $v_i \perp U_0$ . Hence  $\hat{\Lambda}^\top \Lambda_0 \rightarrow \text{diag}(\kappa_i \sqrt{\theta_i \alpha_i})$ , positive diagonal, so the Procrustes rotation  $R^* \rightarrow I$ ; then  $\text{Log}_{[\Lambda_0]}([\hat{\Lambda}]) = \hat{\Lambda} R^* - \Lambda_0 \rightarrow \sum_i [(\kappa_i \sqrt{\theta_i} - \sqrt{\alpha_i}) u_i + \kappa_i \sqrt{1 - \theta_i} v_i] e_i^\top$  and  $B(c)$  is its squared norm; positivity because Cauchy–Schwarz  $2\kappa \sqrt{\theta \alpha} < \alpha + \kappa^2$  is strict unless  $\kappa \theta = \kappa^2 = \alpha$ , impossible for  $c > 0$  ( $\theta < 1$ ).  $\square$

#### A.4 Theorem 4.4

Visible block:  $U_0^\top \widehat{\Lambda}_\star = P_n \text{diag}(\widehat{\beta}_i)$  and  $\Theta^{1/2} \text{diag}(\beta_i) = \Delta^{1/2}$ , so

$$\sqrt{n}(U_0^\top \widehat{\Lambda}_\star - \Delta^{1/2}) = \sqrt{n}(P_n - \Theta^{1/2}) \text{diag}(\beta_i) + \Theta^{1/2} \text{diag}(\sqrt{n}(\widehat{\beta}_i - \beta_i)) + o_P(1).$$

Chain rule:  $\widehat{\beta}_i = \beta(\psi^{-1}(\lambda_{n,i}))$  up to the  $O_P(n^{-1})$  perturbation of  $\widehat{\sigma}_\star^2$  and  $c_n \rightarrow c$  (both Slutsky-negligible; the first by Theorem 2 of [1]), whence  $\sqrt{n}(\widehat{\beta}_i - \beta_i) = \frac{\beta'(\alpha_i)}{\psi'(\alpha_i)} \sqrt{n}(\lambda_{n,i} - \rho_i) + o_P(1)$ . The Procrustes/horizontality correction is, by Lemma 2.2 (whose remainder bound is  $p$ -uniform) applied with  $D_n = O_P(n^{-1/2})$  *restricted to the visible block* – the transverse component of  $\widehat{\Lambda}_\star R^* - \Lambda_0$  is  $O(1)$  but purely horizontal-transverse and does not enter the Sylvester equation, which involves only  $\Lambda_0^\top D = \Delta^{1/2} U_0^\top D$  – exactly the map  $\mathcal{H}_\Delta$  of (18) applied to the displayed limit. Assumption 4.2 and the continuous mapping theorem give (19). For (iii): at  $c = 0$ ,  $\theta \equiv 1$  gives  $\beta = \sqrt{\alpha}$ ,  $\beta' = 1/(2\sqrt{\alpha})$ , and  $\theta\beta'^2\zeta^2/\psi'^2 = 2a^2/(4\alpha) = a^2/(2\alpha)$ , matching (8) (the  $s_0$  term vanishes as  $p \rightarrow \infty$ ); the off-diagonal reduction uses  $\tau_{ij}^2 \rightarrow a_i a_j / (\alpha_i - \alpha_j)^2$  and the cancellation of Appendix A.2.  $\square$

#### A.5 Theorem 4.5

$v^\top \text{Log } e_i = v^\top \widehat{\Lambda}_\star R^* e_i + 0$ ; since  $R^* \rightarrow I$  with  $R^* - I = O_P(n^{-1/2})$  mixing only visible-block quantities, and  $v^\top \widehat{\Lambda}_\star e_j = O_P(n^{-1/2})$  for all  $j$ , the rotation contributes  $O_P(n^{-1})$ ; hence  $\sqrt{n} v^\top \text{Log } e_i = \widehat{\beta}_i \sqrt{n} v^\top \widehat{u}_i + o_P(1) \rightsquigarrow \beta_i N(0, \nu_i^2)$  by Assumption 4.3 and Slutsky. Computing, with  $\bar{\alpha} = \alpha/\sigma^2$ :  $\beta^2 \nu^2 = \frac{\alpha(\alpha + c\sigma^2)}{\alpha^2 - c\sigma^4} \alpha \cdot \frac{\sigma^2 a}{\alpha(\alpha + c\sigma^2)} = \frac{\sigma^2 \alpha a}{\alpha^2 - c\sigma^4} = \frac{\sigma^2 a}{\alpha \psi'(\alpha)}$ , giving (21).  $\square$

#### A.6 Lemma 5.1

On the regular stratum, the quotient of a flat total space by a compact isometric free action has sectional curvature given by O'Neill's formula,  $K([\xi], [\eta]) = \frac{3}{4} \|[\xi, \eta]^V\|^2 / (\dots) \geq 0$ , bounded on compact neighborhoods of  $[\Lambda_0]$  with bounds depending only on  $\Delta$  (the vertical bracket involves  $(\alpha_i + \alpha_j)^{-1}$  factors as in (6)), uniformly in  $p$ . Parallel transport along a geodesic of length  $\ell$  differs from the ambient identification of horizontal spaces by  $O(K\ell^2) = O(n^{-1})$  on bounded vectors.  $\square$

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